

Fly GACA — The Office

<div align="center">

Fly GACA — The Office

The complete operating system for Saudi Arabia's independent civil aviation education platform.

Every document, policy, playbook, and specification that runs Fly GACA — in one version-controlled repository.

 [docs-check](#)  [Status](#)  [License](#)  [Languages](#)  [Sections](#)  [PDFs](#)

 [Master Index](#) ·  [Roadmap](#) ·  [Print Pipeline](#) ·  [Arabic Mirror](#)

</div>

What is this?

ay2m/Office ("The Office") is the internal document repository for [Fly GACA](#) — an independent educational platform and open regulatory library for Saudi civil aviation (GACAR, AIP, charts, ground school, and Captain Adel, the AI flight instructor).

This repository stores every operating document that runs the company: strategy and OKRs, governance rules, legal contracts, finance policies, Saudi compliance bundles (PDPL, ZATCA Phase 2 Fatoora e-invoicing, MISA, Nitaqat), HR frameworks, GTM flight school playbooks & proposals, investor materials, brand assets, and product/engineering specifications.

It contains no application source code — Fly GACA's product software lives in separate dedicated repositories:

Repo	Role & Description
ay2m/Office (this repo)	The business operating system — strategy, governance, legal, finance, KSA compliance (PDPL, ZATCA Phase 2), HR & GTM docs
ay2m/FlyGACA	flygaca.com — the bilingual React 19 + Vite PWA and its Express backend for Cloud Run (target region <code>me-central2</code> , Cloud SQL, Moyasar — not yet deployed), plus the regulatory corpus, 55+ flight tools, and NOAA weather feed
ay2m/Captain-Adel	The AI flight instructor service (captadel.com) + BGE-M3 hybrid dense retrieval engine behind chat, function calling & 138-case evals

Repo	Role & Description
ay2m/FlyGACA-ios	The native SwiftUI app family — shared <code>FlyGACAKit</code> package + ELPT and AIP App Store targets
ay2m/FlyGACA-app	Archived. The retired predecessor of <code>ay2m/FlyGACA</code> , read-only, kept for its 1,005-commit history

Four live repositories plus one archived; this is the internal-docs operating system. Earlier documents in this tree describe a `FlyGACA/...` org and six per-module App Store repos — those paths are legacy redirects to `ay2m/...`, and the six repos never existed.

[!IMPORTANT] Fly GACA is not affiliated with GACA (Saudi General Authority of Civil Aviation). Every document and user-facing surface reinforces one rule: verify against the latest official GACA publication. This platform helps you *find and study* regulation — it never replaces it.

Quick Start

```
# Clone the repository
git clone https://github.com/ay2m/Office.git
cd Office

# Browse the Master Index
open _INDEX.md           # macOS
xdg-open _INDEX.md       # Linux

# Rebuild print-ready PDFs after editing any .md
cd tools/print
npm ci                   # one-time install (uses pre-installed Chromium)
npm run build             # incremental build – re-renders only changed docs
npm run check            # verify front-matter & PDF freshness (CI gate)
```

[!NOTE] The print pipeline uses **Node 20+** and headless **Chromium ≥ 131**. Typography fonts are vendored in `tools/print/fonts/`, enabling 100% offline rendering. Full usage guide: [tools/print/README.md](#).

Repository Structure

The repository is organized into twelve numbered sections (`00–11`), each independently structured and browsable:

Section	Title	Contents & Scope
00	Strategy	Master roadmap, annual strategic plan & OKRs, CEO execution plan (Sprints 0–3), Phase 0 tracker, and strategic brainstorms (00–10)
01	Governance	Founders' agreement, SHA, ESOP plan, Code of Conduct, decision log, board packs, repo governance (<code>CLAUDE.md</code> , <code>CONTRIBUTING.md</code> , <code>SECURITY.md</code> , <code>LICENSE</code>)
02	Legal	NDAs, EULA, SLA, pilot agreement, PDPL privacy policy, DPA, IP & takedown procedure, refund policy, and Saudi lawyer briefs
03	Finance	Banking & treasury policy, procurement, expense policy, budget-vs-actual trackers, financial dashboards, ZATCA e-invoicing templates
04	Compliance (KSA)	ZATCA Phase 2 Fatoora e-invoicing architecture, MISA investment license, PDPL DPIA & compliance program, Nitaqat plan, BCP/DR plans
05	People	Saudi-compliant employment contracts, employee handbook, job descriptions, onboarding/offboarding checklists, performance review templates
06	Operations / IT	Digital office setup, product specs (CRM, Captain Adel, Instructor Dashboard), runbooks, infrastructure diagrams, setup kits
07	Go-To-Market	Flight Academy B2B pilot proposals (GACAR Part 141 ATOs), sales playbooks, demo scripts, objection handling, B2B pipeline, SEO strategy
08	Customer Success	Onboarding playbooks, customer health score framework, NPS survey packs, QBR templates, churn prevention playbooks
09	Investor Relations	Pitch deck, investor FAQ, due diligence questionnaire, monthly investor update templates, risk register, Saudi investor targets
10	Academy & Curriculum	Curriculum map, coverage matrix, PPL mock exams, B2C learner paths, instructor onboarding, cadet welcome packs
11	Brand	Falcon design system, design tokens, style guide, logos, print collateral (EN + RTL AR letterheads, business cards, covers)

Support directories:

Directory	Purpose
<code>_INDEX.md</code>	Readable master index across all 12 sections
<code>templates/</code>	Reusable Markdown starters (<code>tpl-fin-report</code> , <code>tpl-hr-policy</code> , <code>tpl-legal-memo</code> , <code>tpl-ops-runbook</code> , <code>tpl-strat-proposal</code>)
<code>ar/</code>	Full Arabic (Saudi MSA) mirror of all 12 sections and templates — identical structure and ASCII filenames
<code>tools/print/</code>	Markdown & HTML → branded A4 PDF render pipeline (Falcon document theme, offline fonts, EN + RTL AR)
<code>_print/</code>	Generated print-ready PDFs mirroring the exact file tree (tracked in git)
<code>contracts/</code>	<code>flygaca-family.json</code> — the cross-repo family contract shared with both product repos (details)

Directory	Purpose
<code>tools/contracts/</code>	<code>stamp-manifest.mjs</code> — stamps and verifies the manifest's self-hash
<code>.claude/agents/</code>	Four project subagents — <code>doc-smith</code> , <code>ar-mirror</code> , <code>ksa-compliance</code> , <code>family-warden</code> . Roster: <code>.claude/agents/README.md</code> ; rationale: <code>06-operations-it/agent-workforce-plan.md</code>

🌟 Key Capabilities

Capability	Detail
 Version-Controlled Operations	Every policy, contract, and technical spec is tracked in Git — full diff history and blame.
 Automated Print Pipeline	Any <code>.md</code> edit re-renders to a branded, watermarked A4 PDF via standard headless Chromium.
 Bilingual Native (EN / AR)	Parallel Arabic (Saudi MSA) mirror under <code>ar/</code> ; ASCII kebab-case filenames remain identical for diffing (128 EN / 128 AR).
 YAML Front-Matter Metadata	Standardized <code>title / section / doc_type / status / owner / last_updated / lang</code> on every document.
 KSA Regulatory Stack	Purpose-built for Saudi legal frameworks: PDPL, ZATCA Phase 2 Fatoora e-invoicing, MISA licensing, and Nitaqat.
 Captain Adel Technical Specs	Full AI flight instructor specs, refusal protocol, deployment runbooks, and evaluation suite.
 Falcon Design System	Inter body font, Cairo headings & Arabic text, JetBrains Mono code, and Falcon Blue accent palette.

🌐 Arabic Localization (`ar/`)

The `ar/` directory is a complete, parallel Arabic translation of the document tree — same folder structure, same ASCII filenames, written in **Modern Standard Arabic (Saudi official register)**.

- 128 `.md` files maintained in 100% sync with the English source
- Unified terminology using `ar/_GLOSSARY.md` (EN↔AR term glossary)
- Arabic PDFs automatically render **right-to-left** (Cairo typography, RTL margin boxes)
- Latin code paths, identifiers, and file names remain LTR inside Arabic text

[!NOTE] English is the authoritative source. In case of any ambiguity, the English document governs.

Print Pipeline & CI Gate

Every `.md` document generates a **branded, print-ready PDF** under `_print/`, mirroring the folder tree. The pipeline uses `markdown-it` + headless Chromium with the Falcon document theme:

- **Fonts:** Inter (body) · Cairo (headings + Arabic) · JetBrains Mono (code) — vendored offline under `tools/print/fonts/`
- **Layout:** A4, 0.75 in margins, footer page numbers, cover block generated from YAML front-matter
- **Watermarks:** `status: draft` or `scaffold` automatically stamps a DRAFT/SCAFFOLD watermark
- **Incremental:** `.buildcache.json` tracks content hashes — unchanged docs are skipped

```
cd tools/print
npm run build          # incremental rebuild
npm run build:force    # rebuild all PDFs from scratch
node build.mjs 02-legal/terms-of-use-draft-2026-06-14.md # render a single document
node check.mjs      # run CI freshness & front-matter validation gate
node check-facts.mjs # the entity-facts gate — see The Family Contract below
```

The Family Contract

The three active repositories share one versioned artifact: `contracts/flygaca-family.json`, committed **byte-identically** to this repo, `ay2m/FlyGACA` and `ay2m/Captain-Adel`. It exists because the family's cross-repo claims used to live only in prose and drifted with nothing failing.

Three blocks, each naming the repo that **owns** it — only the owner edits its block:

Block	Owner	Source of truth	Mirrored into
<code>entity</code>	this repo	<code>01-governance/company-facts.md</code>	<code>ay2m/FlyGACA</code> 's <code>jsonld.ts</code> + both i18n bundles; <code>ay2m/Captain-Adel</code> 's <code>footer.js</code> , <code>terms.html</code> , <code>privacy.html</code> , <code>package.json</code> , <code>LICENSE</code>
<code>chat</code>	<code>ay2m/FlyGACA</code>	its <code>server/src/contract.ts</code>	the answer shape both Captain Adel implementations must honour
<code>repos</code>	this repo	the roster above	supersedes any prose citing a <code>FlyGACA/...</code> org

Each repo gates its own half in the CI it already runs. Here that is `node tools/print/check-facts.mjs` (wired into `docs-check.yml`, aliased `npm run check:facts`), which asserts every `entity` value against the `company-facts.md` table it was copied from — and asserts the **IBAN** and

account number from that same document are absent from the manifest, since the manifest travels to both product repos. The product repos run `tests/family-contract.test.ts` and `test/family-contract.test.js`.

Changing a company fact is a three-repo change. Edit `company-facts.md`, update the manifest, bump its `version`, re-stamp its hash with `node tools/contracts/stamp-manifest.mjs contracts/flygaca-family.json`, copy it verbatim into both product repos, and open all three PRs together. Every repo's gate stays red until its half is done.

Known limitation: nothing offline can prove the three copies are the same revision. `version` and `sha` reduce that to a visible one-line diff; closing it fully needs a scheduled cross-repo workflow, which does not exist yet.

Brand Identity

The Falcon design system, logo assets, and print templates in `11-brand/` give every Fly GACA document a uniform identity.

<p align="center"> </p>

Roadmap & Execution Status

Phases tracked in `00-strategy/roadmap.md`.

Phase	Title	Status	Summary
0	Foundations	 In progress	Technical stack live; Saudi legal entity registration open
1	Open Library	 Shipped	GACAR regulations, aerodromes, and chart index live
2	Captain Adel AI	 Shipped	RAG flight instructor live on Gemini 2.5 Flash
3	Pilot Accounts	 Built	Session-cookie auth + Google OAuth on Cloud Run, Cloud SQL Postgres live; PDPL DPIA pending
4	Arabic & Polish	 In progress	Bilingual engine live; inner page translations underway
5	Monetization & Schools	 Planned	Gated on legal entity registration
6	Native Apps	 In progress	iOS ELPT & AIP targets built; store submission pending

Phase	Title	Status	Summary
7	Training Platform	✅ Shipped	162-question bank, flashcards, and exam analytics live
8	Library Platform	✅ Shipped	Whole-library search, safety lessons, and reading paths live
9	Launch & Visibility	🟡 Deployed	Web app deployed; custom domain & GTM active
10	Captain Adel Prod	🟡 In progress	Input guards, rate limits & evals shipped
11	Practical Depth	✅ Shipped	21 flight tools, 11 guides, full study bank complete

🤝 Governance & Contributing

This repository strictly follows the governance guidelines in [01-governance/](#):

- [CONTRIBUTING.md](#) — setup, doc conventions, pre-PR checklist
- [CODE_OF_CONDUCT.md](#) — community standards
- [SECURITY.md](#) — security & responsible disclosure policy
- [CLAUDE.md](#) — AI assistant guidelines and architecture documentation

All custom operational documents are licensed under [Apache License 2.0](#). Regulatory texts quoted belong to GACA and are excluded from this license.

<div align="center">

Built for Saudi skies. Operated from The Office.

</div>